



Philip Hansett
Site Vice President
225-381-4374

RBG-48248

10 CFR 50.73

September 13, 2023

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Subject: Licensee Event Report 50-458 / 2023-002-01, Division I and II Diesel
Generators Inoperable due to Exceeding Load Sequence Times

River Bend Station – Unit 1
NRC Docket Nos. 50-458
Renewed Facility Operating License No. NPF-47

On June 29, 2023, Licensee Event Report 50-458 / 2023-002-00, Division I and II Diesel Generators Inoperable due to Exceeding Load Sequence Times (ML23180A213) was submitted by Entergy Nuclear Operating Inc. This letter is a supplement to the report to provide the updated results of the causal evaluation. The updated information is denoted by revision bars located in the right-hand margin. The report is submitted in accordance with 10 CFR 50.73.

This document contains no commitments.

Should you have any questions, please contact Mr. Randy Crawford, Regulatory Assurance Manager, at 225-381-4177.

Respectfully,

A handwritten signature in black ink, appearing to read 'Phil Hansett', with a long horizontal flourish extending to the right.

Phil Hansett

PH/twf

Enclosure: Licensee Event Report 50-458 / 2023-002-01, Division I and II Diesel
Generators Inoperable due to Exceeding Load Sequence Times

cc: NRC Region IV Regional Administrator - Region IV
NRC Senior Resident Inspector - River Bend Station

Enclosure

RBG-48248

**Licensee Event Report 50-458 / 2023-002-01, Division I and II Diesel Generators
Inoperable due to Exceeding Load Sequence Times**

NRC FORM 366 (09-01-2023)		U.S. NUCLEAR REGULATORY COMMISSION		APPROVED BY OMB: NO. 3150-0104		EXPIRES: 09/30/2023						
		LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)				Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov , and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: pira_submission@omb.eop.gov . The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.						
1. Facility Name River Bend Station, Unit 1				☒ 050 ☐ 052		2. Docket Number 458		3. Page 1 OF 3				
4. Title Division I and II Diesel Generators Inoperable due to Exceeding Load Sequence Times												
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved			
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name N/A	☐ 050 Docket Number N/A		
02	12	2023	2023	- 002 -	01	09	13	2023	Facility Name N/A	☐ 052 Docket Number N/A		
9. Operating Mode 4						10. Power Level 0%						
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)												
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)		
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)		
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)		
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)		
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☒ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)		
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)		☐ 73.1200(f)		
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)		
☐ 20.2203(a)(2)(iv)				☒ 50.73(a)(2)(i)(B)		☒ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)		
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)						
☐ OTHER (Specify here, in abstract, or NRC 366A).												
12. Licensee Contact for this LER												
Licensee Contact Randy Crawford								Phone Number (Include area code) 225-381-4177				
13. Complete One Line for each Component Failure Described in this Report												
Cause	System	Component	Manufacturer	Reportable to IRIS		Cause	System	Component	Manufacturer	Reportable to IRIS		
X	EK	N/A	N/A	N/A		N/A	N/A	N/A	N/A	N/A		
14. Supplemental Report Expected						15. Expected Submission Date				Month	Day	Year
☒ No		☐ Yes (If yes, complete 15. Expected Submission Date)								N/A	N/A	N/A
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced type written lines)												
On February 12, 2023, with River Bend Station (RBS) in Mode 4 during Refueling Outage 22, testing of the Division II Emergency Core Cooling System (ECCS) revealed that both Division II Control Building Chillers (HVK-CHL1B and HVK-CHL1D) failed to sequence onto the emergency bus within the Technical Specification (TS) required time. Investigation of the issue revealed that the same condition was present for the Division I Control Building Chillers (HVK-CHL1A and HVK-CHL1C). The condition described above was caused by a modification that replaced the Control Building Chiller controllers. The preliminary root cause identified that when the controllers were replaced, a delay was introduced in the start logic. Corrective actions to restore TS compliance were completed prior to the next mode of applicability. This condition resulted in the inability to comply with a Surveillance Requirement of TS 3.8.1. At the time of discovery, TS 3.8.1 was not applicable. An evaluation determined that the TS 3.8.1 Surveillance Requirement was not met prior to time of discovery and Required Actions were not taken within the associated allowed Completion Times in accordance with TS 3.8.1 - AC Sources – Operating.												

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: aira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME River Bend Station, Unit 1	☒ 050	2. DOCKET NUMBER 458	3. LER NUMBER		
	☐ 052		YEAR 2023	SEQUENTIAL NUMBER - 002	REV NO. - 01

NARRATIVE**EVENT DESCRIPTION**

On February 12, 2023, RBS performed STP-309-0602, DIVISION II ECCS TEST. During the performance of Section 7.4, ECCS/LOP/LOCA Test Initiation, HVK-CHL1B [KM] did not load sequence within the required time. Load sequence time was 237.1 seconds with acceptance criteria of 180.9 to 221.1 seconds. During performance of Section 7.25, ECCS/LOP Test Initiation, HVK-CHL1D [KM] did not load sequence within the required time. Load sequence time was 241 seconds with acceptance criteria of 180.9 to 221.1 seconds. Condition Reports were generated for both issues.

On May 2, 2023, the station determined that the Division I and Division II Diesel Generators [EK] had been inoperable as a result of the chiller controller modification. HVK-CHL1C [KM] and HVK-CHL1D modifications were completed in 2014. HVK-CHL1A [KM] and HVK-CHL1B modifications were completed in 2020.

The discovery date was determined to be February 12, 2023. The event is reportable as an Operation or Condition Prohibited by Technical Specifications per 10 CFR 50.73(a)(2)(i)(B) and an Event or Condition that Could Have Prevented Fulfillment of a Safety Function per 10 CFR 50.73(a)(2)(v)(A) and (D).

EVENT CAUSE

This event was caused by an error that was introduced during a modification to the HVK-CHL1(A)(B)(C)(D) control logic. Beginning in 2014, the analog control logic was modified to digital control logic. A bypass timer in the analog logic was not replicated in the new digital logic. The bypass timer blocked a low chiller condenser water flow signal for a short duration during the chiller start sequence. If received, the low water flow signal causes the chiller start sequence to reset. Because the signal was no longer blocked in the digital control logic, the low chiller condenser flow signal was capable of causing the chiller start sequence to reset resulting in an extended chiller start sequence.

SAFETY ASSESSMENT

The Division I and II Diesel Generators were declared inoperable based on load sequence time of the HVK Chillers exceeding TS SR 3.8.1.18 requirements. The following engineering analysis provided evidence that the delay had no effect on the generator loading sequence.

Calculation E-192, STANDBY DIESEL GENERATOR LOADING CALCULATION, evaluates the loading of the standby diesel generators during the worst-case loading conditions (LOP concurrent with LOCA). The calculation verifies that the maximum magnitude for the load blocks is within the tested values of loading established by the vendor. HVK-CHL1B/D are in the 1.5-10-minute load block for the Division II Diesel Generator. The chiller is the last component in the load block to start with a listed start time of 211 seconds from the LOP. The next component to start is at the 10-minute mark. HVK-CHL1A/C are in the 1.5-10-minute load block for the Division I Diesel Generator. The chiller is the last component to start in the 1.5-10-minute load block with a start time of 241 seconds. Even with the additional 40 second delay, the chiller will remain in the 1.5-10-minute load block and there will be at least 5 minutes between a chiller start and the start of the next component. Therefore, the delay has no effect on the generator loading sequence.

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NARRATIVE

Based on the engineering analysis, the Division I and II Diesel Generators remained capable of performing their required safety function for the duration of the event and the event will not be counted as a Safety System Functional Failure (SSFF) for the NRC Performance Indicators. The analysis also supports that there were no consequences to the health and safety of the public.

ADDITIONAL INFORMATION

The Control Building [NA] Air Conditioning System (CBAC) consists of two independent, redundant subsystems that provide cooling and heating of the control building air. The associated control building chilled water subsystem (HVK) supplies chilled water to both subsystem AC units, as well as to the Main Control Room AC units.

The design of the AC electrical power system provides independence and redundancy to ensure an available source of power to the Engineered Safety Feature (ESF) systems. Each ESF bus has a dedicated onsite Diesel Generator (DG). A DG starts automatically on loss of coolant accident (LOCA) or an ESF bus degraded voltage, undervoltage/loss of power (LOP) signal. Certain required plant loads in Divisions I and II (including HVK) are returned to service in a predetermined sequence to prevent overloading the DG.

RBS completed controller modifications for HVK-CHL1C and HVK-CHL1D in 2014. HVK-CHL1A and HVK-CHL1B modifications were completed in 2020. A review of historical chiller data logs shows that since the HVK chiller controls upgrade, the start of HVK chillers may be delayed past the design start time by approximately 40 seconds.

TS Surveillance Requirement (SR) 3.8.1.18 states, "Verify sequence time is within +/- 10% of design for each load sequencer timer."

CORRECTIVE ACTIONS**Completed Corrective Actions**

1. Engineering Change Modifications were completed for all Control Building Chillers to correct sequence delays as detailed below:
 - Work Order 594333-01 Modify HVK-CHL1A Start Logic per EC-95152 - COMPLETE
 - Work Order 594334-01 Modify HVK-CHL1B Start Logic per EC-95151 - COMPLETE
 - Work Order 594335-01 Modify HVK-CHL1C Start Logic per EC-95153 - COMPLETE
 - Work Order 594336-01 Modify HVK-CHL1D Start Logic per EC-95150 - COMPLETE

PREVIOUS SIMILAR OCCURRENCES

None.

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX]. River Bend equipment codes are identified as (XX).